FOOD EMPATHY

Connecting Surplus Food with People in Need

A smart map-based food-sharing platform

CORE IDEA

Connect donors, NGOs, volunteers and communities through one location-aware platform.

DONATE → CONNECT → DELIVER



Student / Team Name
College Name

The Problem

Food waste and food insecurity exist side by side.

SURPLUS FOOD

Restaurants, events, hotels, supermarkets and house

LOCATION GAP

Donors may not know where nearby food is needed

COORDINATION GAP

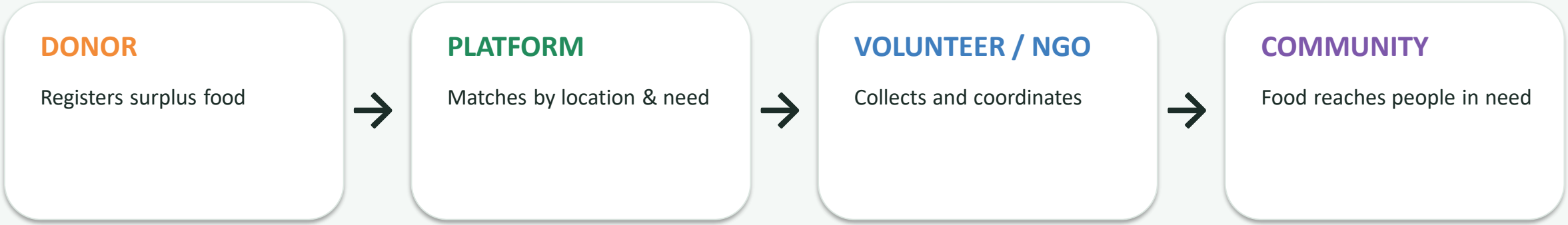
NGOs and volunteers need faster communication, routin

FOOD WASTE → **LOCATION + TIME** → **FOOD NEED**

A real-time, map-based coordination layer can reduce the gap.

Our Solution

One platform connecting food, people and places.



- Donation registration
- Interactive map
- Nearby needs & collection points
- Pickup/delivery tracking
- Notifications and donation history

Map-Based Food Assistance

The map is the central coordination layer.



MAP FEATURES

Current location

Nearby donors & NGOs

Food-need zones

Route navigation

Distance & ETA

Map filters

How the System Works

From surplus food to completed distribution.

01 Register surplus food → **02** Verify food details → **03** Add location to map → **04** Notify nearby NGO / volunteer

05 Accept pickup → **06** Collect food → **07** Deliver / distribute → **08** Mark completed

User Roles

Each role has a focused workflow.

DONOR

Add surplus food

Quantity & expiry

Pickup location

Track status

NGO / FOOD ORG

Post requirements

Accept donations

Manage distribution

Monitor requests

VOLUNTEER

View nearby tasks

Accept pickup

Navigate routes

Update delivery

ADMIN

Verify users/orgs

Monitor donations

Manage locations

Handle reports

Smart Map Features

Location intelligence turns coordination into action.

Current location

Location-aware discovery and action.

Nearby food donors

Location-aware discovery and action.

Nearby NGOs

Location-aware discovery and action.

Food collection centres

Location-aware discovery and action.

Food requirement areas

Location-aware discovery and action.

Route navigation

Location-aware discovery and action.

Distance calculation

Location-aware discovery and action.

Estimated travel time

Location-aware discovery and action.

Location-based notifications

Location-aware discovery and action.

Map filters

Location-aware discovery and action.

Technology Stack

Modular architecture for a scalable prototype.

FRONTEND

Flutter / React

Responsive UI

Map interface

BACKEND

FastAPI / Node.js

REST APIs

Authentication

Donation management

DATABASE

PostgreSQL

Users

Donations

Locations

Requests

MAPS

Google Maps API or OpenStreetMap

Geolocation

Routing

Distance

Mobile / Web App → **API Server** → **PostgreSQL**

Maps API ↗

Notification Service

Empathy & Social Impact

Technology is used to strengthen community action.

ENVIRONMENTAL

Reduce avoidable food waste

Encourage responsible sharing

SOCIAL

Connect surplus food with people who need assistar

Support NGOs and volunteers

COMMUNITY

Build a local food-sharing network

Improve coordination and visibility

SURPLUS FOOD + TECHNOLOGY + COMMUNITY = IMPACT

Safety & Food Quality

Responsible donation needs clear information and oversight.

- Food type and preparation information
- Use-by / expiry information
- Quantity and storage condition
- Donor and organization verification
- Report inappropriate or unsafe donations
- Admin monitoring and audit trail

IMPORTANT

The platform should encourage compliance with applicable local food-safety re

Future Scope

Ideas that can make the platform more intelligent and scalable.

- AI-based food demand prediction
- Smart donor–NGO matching
- AI-assisted route optimization
- Food-waste prediction
- Real-time delivery tracking
- Multilingual and voice support
- Food-demand heat maps
- Corporate and restaurant partnerships
- Analytics for impact measurement

FOOD EMPATHY

Don't Waste Food. Share It With Empathy.

DONATE → CONNECT → COLLECT → DELIVER → IMPACT

A map-based platform for more organized, transparent and location-aware food sharing.

Thank You

Team Name • College Name • Contact